

TSUIKI AB, Japan

WMO: 478400

Lat: 33.685N

Lon: 131.040E

Elev: 55

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.4	30.2	17.6	13.4	38.4	19.7	14.8	38.5	25.5	40.8	23.0	42.4	7.8	240	0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.9	91.2	78.2	89.3	78.1	87.5	78.1	80.1	86.8	79.4	86.1	78.6	85.2	11.4	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.7	149.3	84.5	77.3	142.1	83.5	76.7	139.6	83.1	44.0	87.5	43.0	86.1	42.1	85.7	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.1	20.3	18.0	DB	24.8	94.2	1.7	1.9	23.6	95.6	22.6	96.7	21.7	97.7	20.4	99.1	
			WB	22.9	81.7	1.4	1.2	21.9	82.5	21.0	83.2	20.3	83.9	19.2	84.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.2	40.9	42.1	47.6	56.1	64.7	71.8	79.4	81.0	74.3	64.4	54.1	45.0
	DBStd	14.65	4.34	4.83	5.37	5.46	4.22	3.97	3.87	3.04	4.49	4.92	5.68	5.11
	HDD50	828	283	227	112	12	0	0	0	0	0	0	23	171
	HDD65	3275	746	642	540	270	57	1	0	0	0	70	329	620
	CDD50	4556	2	5	37	195	456	655	910	961	729	445	146	15
	CDD65	1529	0	0	0	3	48	206	445	496	279	50	2	0
	CDH74	12564	0	0	2	23	196	910	4156	5208	1899	167	3	0
	CDH80	3865	0	0	0	3	15	135	1379	1924	401	8	0	0
Wind	WSAvg	7.8	8.5	8.1	8.5	7.9	7.6	7.2	6.8	7.7	7.8	7.6	7.5	8.3
Precipitation	PrecAvg	76.8	2.4	3.2	4.8	5.7	6.6	14.5	12.5	7.8	8.8	4.7	3.5	2.4
	PrecMax	122.1	7.1	6.2	13.4	10.1	12.0	27.2	27.2	15.3	20.0	14.2	7.7	5.0
	PrecMin	46.0	0.7	1.1	1.8	1.6	2.2	2.8	2.8	2.1	1.9	0.6	0.9	0.2
	PrecStd	15.0	1.5	1.3	2.2	2.1	2.5	6.1	7.0	4.1	5.2	3.5	1.8	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.8	62.3	69.7	77.0	82.2	87.6	93.3	93.6	89.6	80.9	71.6	62.7
		MCWB	50.5	56.1	58.3	61.7	64.1	76.3	77.6	78.1	77.7	71.2	63.1	56.1
	2%	DB	53.9	57.1	64.2	71.7	78.6	83.8	91.1	91.4	86.2	78.4	68.4	59.3
		MCWB	47.2	49.5	53.5	58.6	63.5	73.3	77.9	78.5	76.9	69.7	61.3	52.5
	5%	DB	51.8	53.8	60.6	68.3	75.6	80.9	88.2	89.6	84.2	75.4	66.4	57.2
		MCWB	45.6	47.0	51.7	57.8	63.4	71.8	77.9	78.3	75.4	67.3	60.0	50.9
	10%	DB	49.8	51.6	57.3	66.1	73.6	78.9	86.4	87.9	82.3	73.5	64.4	55.1
		MCWB	44.5	45.7	50.3	56.8	63.1	70.9	77.5	78.5	74.0	65.6	58.4	49.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.4	56.7	59.9	66.6	70.4	78.0	81.3	81.8	80.1	74.6	67.1	58.4
		MCDB	55.0	61.0	65.8	71.0	74.7	83.7	86.9	87.7	85.7	78.4	69.8	61.0
	2%	WB	49.6	52.1	56.0	63.0	68.4	76.1	79.9	80.6	78.4	71.8	63.7	54.8
		MCDB	51.9	55.3	60.8	67.9	73.0	81.2	86.6	87.4	83.7	75.9	66.8	58.0
	5%	WB	47.0	49.4	53.6	60.8	66.9	74.5	79.1	79.8	77.1	69.3	61.7	52.4
		MCDB	50.0	51.8	58.7	65.7	71.9	79.0	85.9	86.6	82.0	73.8	65.0	55.8
	10%	WB	44.7	46.9	51.5	59.2	65.4	72.8	78.2	79.1	75.8	67.3	59.5	50.3
		MCDB	48.4	50.0	56.0	64.0	70.8	77.1	84.8	85.8	80.6	71.8	63.3	53.7
Mean Daily Temperature Range	5% DB	MDBR	14.8	15.6	16.6	17.4	16.3	12.0	11.7	12.9	13.1	15.5	16.2	15.5
		MCDBR	17.8	19.1	21.3	21.2	20.1	15.2	14.9	15.1	14.3	16.6	17.2	18.2
		MCWBR	12.4	13.5	13.6	12.3	11.2	7.6	6.6	6.6	7.4	9.7	11.5	12.9
	5% WB	MCDBR	16.1	15.7	17.9	16.1	14.5	11.0	13.0	13.3	12.1	13.3	14.3	15.6
		MCWBR	13.1	13.3	13.3	11.0	9.7	6.5	6.7	6.9	7.1	9.0	11.1	12.5
Clear-Sky Solar Irradiance	taub	0.427	0.469	0.531	0.552	0.558	0.571	0.570	0.535	0.504	0.459	0.430	0.420	
	taud	2.065	1.991	1.843	1.846	1.893	1.944	1.991	2.083	2.126	2.159	2.151	2.103	
	Ebn at Noon	238	241	237	239	239	236	235	241	242	243	236	232	
	Edh at Noon	42	50	62	65	63	60	56	51	47	42	38	38	
All-Sky Solar Radiation	RadAvg	699	922	1264	1525	1654	1410	1542	1620	1291	1089	814	645	
	RadStd	63	87	86	135	187	168	227	174	136	114	75	43	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.59	N/A	N/A	N/A	-0.83	-1.24	N/A	N/A	N/A	N/A
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page